

RAM CHARAN VANGA

RA2511027010164

Subject : DSA

Unit 2, S6, SLO1 - Singly Linked List – Insertion at the Middle

Algorithm to insert a new node VAL after a node that has value NUM

Step 1: IF AVAIL = NULL

Write OVERFLOW

Go to Step 12

[END OF IF]

Step 2: SET NEW_NODE = _____

Step 3: SET AVAIL = _____

Step 4: SET NEW_NODE -> DATA = _____

Step 5: SET PTR = _____

Step 6: SET PREPTR = _____

Step 7: Repeat Steps 8 and 9 while _____ != NUM

Step 8: SET PREPTR = _____

Step 9: SET PTR = _____

[END OF LOOP]

Step 10: _____ = NEW_NODE

Step 11: SET NEW_NODE -> NEXT = _____

Step 12: EXIT

Answer: Step 2: SET NEW_NODE = AVAIL

Step 3: SET AVAIL = AVAIL -> NEXT

Step 4: SET NEW_NODE -> DATA = VAL

Step 5: SET PTR = START

Step 6: SET PREPTR = PTR

Step 7: while PREPTR -> DATA != NUM

Step 8: SET PREPTR = PTR

Step 9: SET PTR = PTR -> NEXT

Step 10: PREPTR -> NEXT = NEW_NODE

Step 11: SET NEW_NODE -> NEXT = PTR

Step 12: EXIT

Algorithm to insert a new node VAL before a node that has value NUM

Step 1: IF AVAIL = NULL

Write OVERFLOW

Go to Step 12

[END OF IF]

Step 2: SET NEW_NODE = AVAIL

Step 3: SET AVAIL = AVAIL -> NEXT

Step 4: SET _____ = VAL

Step 5: SET _____ = START

Step 6: SET _____ = PTR

Step 7: Repeat Steps 8 and 9 while _____ != NUM

Step 8: SET _____ = PTR

Step 9: SET PTR = _____

[END OF LOOP]

Step 10: SET _____ = NEW_NODE

Step 11: SET NEW_NODE -> NEXT = _____

Step 12: EXIT

Answer: Step 4: SET NEW_NODE -> DATA = VAL

Step 5: SET PTR = START

Step 6: SET PREPTR = PTR

Step 7: while PTR -> DATA != NUM

Step 8: SET PREPTR = PTR

Step 9: SET PTR = PTR -> NEXT

Step 10: SET PREPTR -> NEXT = NEW_NODE

Step 11: SET NEW_NODE -> NEXT = PTR

Step 12: EXIT